
Physically-Aware Hierarchical GaLore for Memory-Efficient SciML Training

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Abstract

1 Training billion-parameter scientific machine learning (SciML) models on com-
2 modity GPUs is impeded by optimizer-state memory and by the need to preserve
3 physical constraints. Gradient low-rank projection, exemplified by GaLore, reduces
4 optimizer memory by projecting gradients into low-dimensional subspaces while
5 keeping all weights trainable, but suffers from costly full-matrix SVD refreshes,
6 physics-agnostic subspaces, and limited evaluation beyond NLP Zhao et al. [2024].
7 We introduce Physically-Aware Hierarchical GaLore (Phi-GaLore), a two-stage gra-
8 dient projection framework for SciML. Stage 1 maintains an exponential-moving-
9 average randomized sketch of each large gradient using CountSketch on rows
10 and columns to form a tiny $r \times r$ core in $\mathcal{O}((m+n)r)$ time per step. Stage 2
11 periodically performs an SVD on the $r \times r$ covariance of this core and augments the
12 resulting learned subspace with analytic physics bases (e.g., divergence-free and
13 SO(3)-aligned blocks), followed by orthonormalization and cross-layer projector
14 sharing. This design eliminates large SVDs, encodes domain symmetries, and
15 amortizes cost across layers, and is implemented as a drop-in PyTorch optimizer
16 wrapper compatible with 8-bit states and FLORA-style compression Hao et al.
17 [2024]. In instrumented, short-horizon PyTorch studies, Phi-GaLore achieves
18 near-perfect cosine alignment with full gradients at rank 32, reduces divergence
19 norms on a toy incompressible Navier-Stokes task when injecting a divergence-free
20 basis, and lowers per-step overhead via hierarchical sharing. While these brief
21 runs do not demonstrate steady-state wall-clock gains over GaLore or quantify
22 memory savings at scale, the observed behaviors validate the core mechanisms and
23 motivate longer SciML evaluations on PINNsFormer, Transolver, and molecular
24 transformers Zhao et al. [2023], Wu et al. [2024], Chen et al. [2024], Alkin et al.
25 [2024], Bryutkin et al. [2024].

26 1 Introduction

27 1.1 Context and challenges in scaling SciML

28 Scientific machine learning increasingly relies on large neural operators and molecular transform-
29 ers to approximate complex physical systems, from incompressible flow to long-range molecular
30 interactions. Recent architectures transfer the advantages of transformers to PDEs and geometry-rich
31 domains: PINNsFormer recasts physics-informed learning as sequence modeling Zhao et al. [2023],
32 Transolver designs physics-aware attention for complex geometries Wu et al. [2024], position-induced
33 transformers encode spatial relations efficiently Chen and Wu [2024], and universal physics trans-
34 formers unify operator learning across meshes and particles Alkin et al. [2024]. Graph transformer
35 neural operators broaden applicability to arbitrary geometries Bryutkin et al. [2024], while molecular
36 models emphasize invariance and equivariance Chen et al. [2024], Luo et al. [2022], Rong et al.
37 [2020]. Scaling these models to billions of parameters under realistic GPU budgets is constrained by

optimizer-state memory and by the need to respect symmetries and conservation laws throughout training.

1.2 Motivation for physics-aware low-rank gradient projection

Gradient low-rank projection, as in GaLore, offers a way to reduce optimizer-state memory while preserving full-parameter updates Zhao et al. [2024]. GaLore periodically performs full-matrix SVD on large gradients to construct low-rank projectors, demonstrating strong memory savings for language models. However, three obstacles limit direct adoption in SciML. First, periodic full-matrix SVDs are expensive: the wall-clock overhead grows quadratically with hidden size, which is non-negligible for scientific transformers with wide layers. Second, the data-driven subspaces learned by GaLore are agnostic to physical symmetries, risking the removal of gradient directions essential for incompressibility, rotational equivariance, or causality. Third, generality outside NLP remains uncertain: whether low-rank gradient structure persists in stiff PDEs or long-range molecular tasks is underexplored.

Beyond practical concerns, evidence from randomized sketching and optimization theory suggests that gradients and second-moment matrices often concentrate on small subspaces, motivating sketch-based tracking of leading directions Feinberg et al. [2023]. Analyses of matrix sensing and factorization further indicate incremental low-rank learning dynamics Jin et al. [2023], Li et al. [2020]. These insights, together with the symmetry requirements of SciML, motivate a modified formulation of gradient projection that is both efficient and physics-aware.

1.3 Overview of the approach

We propose Physically-Aware Hierarchical GaLore (Phi-GaLore), a two-stage, symmetry-aware gradient projection method tailored to SciML. Stage 1 replaces full SVDs with a fast per-step randomized sketch, building and maintaining an exponential-moving-average $r \times r$ core for each group of layers. Stage 2 occasionally performs a tiny SVD on this core, lifts the learned directions back to row space, and augments them with analytic physics bases such as divergence-free Fourier modes (for velocity outputs) or SO(3)-aligned blocks (for equivariant mixing). A two-pass orthonormalization yields the final projector. To reduce overhead in deep models, we introduce hierarchical projector sharing across layers at similar resolution or function. We implement Phi-GaLore as a drop-in optimizer wrapper compatible with 8-bit states and with FLORA-style momentum compression Hao et al. [2024].

We validate the core mechanisms through three short, controlled experiments in PyTorch: a micro-benchmark measuring gradient alignment and projection overhead on an MLPField and a tiny transformer; a toy incompressible Navier-Stokes experiment testing a divergence-free basis at the output head; and a toy molecular regression probing hierarchical sharing and robustness under rotations and coordinate noise. The brief runs show near-perfect cosine alignment to full gradients at rank 32, substantial reduction in divergence norm when injecting a divergence-free basis, and a roughly 30% reduction in per-step overhead from hierarchical sharing. Because the runs are very short, GaLore’s initial SVD cost dominates only the first step and its periodic refresh cost does not recur, so steady-state wall-clock advantages and memory savings cannot yet be established.

1.4 Contributions

- **Two-stage physics-aware projection:** A symmetry-aware, two-stage gradient projection that replaces full-matrix SVDs with a tiny $r \times r$ SVD on randomized EMA sketches and augments learned subspaces with analytic physics bases.
- **Hierarchical sharing:** A hierarchical sharing mechanism that groups layers at similar resolution to amortize sketching and tiny SVD computations across deep architectures.
- **Practical optimizer wrapper:** A drop-in optimizer wrapper that projects gradients pre-update, supports 8-bit states, and inter-operates with FLORA-style momentum compression Hao et al. [2024], while selectively bypassing small matrices.
- **Instrumented verification:** An instrumented experimental suite demonstrating high cosine alignment to full gradients, divergence-norm reduction via physics-aware bases, and sharing-induced overhead cuts, with released artifacts.

89 1.5 Future directions

90 Looking forward, we plan long-horizon studies and billion-parameter SciML evaluations under fixed
91 GPU budgets to quantify memory savings, throughput, and accuracy; to examine symmetry-aware
92 subspaces in stiff PDE regimes and in molecular transformers; and to compare against GaLore and
93 FLORA at scale Zhao et al. [2024], Hao et al. [2024], Zhao et al. [2023], Wu et al. [2024], Chen et al.
94 [2024], Alkin et al. [2024], Bryutkin et al. [2024].

95 2 Related Work

96 Low-rank gradient projection. GaLore shows that projecting gradients into low-rank subspaces
97 can reduce optimizer-state memory while keeping all parameters trainable, achieving strong results
98 for LLM pretraining and fine-tuning Zhao et al. [2024]. Its projectors are derived from periodic
99 full SVDs of large gradients, a choice that is expensive for wide SciML layers and agnostic to
100 physics. Phi-GaLore follows GaLore’s spirit but replaces the full SVDs with a tiny $r \times r$ SVD on
101 an exponential-moving-average randomized sketch and augments the learned subspace with physics
102 bases, thereby targeting both efficiency and symmetry preservation.

103 Parameter-efficient and gradient-compression methods. LoRA and its extensions train low-rank
104 adapters instead of updating full weights; strong variants combine low-rank with quantization (LQ-
105 LoRA), decompose weights (DoRA), resample updates (ReLoRA), or leverage layerwise sampling
106 (LISA) Guo et al. [2023], Liu et al. [2024], Lialin et al. [2023], Pan et al. [2024], Biderman et al.
107 [2024]. These approaches reduce memory and can accelerate training but constrain updates to
108 low-rank forms and alter training dynamics, which may be problematic when strict physics fidelity is
109 needed. FLORA reframes adapters as gradient compressors via random projections with resampling,
110 achieving high-rank effective updates with sublinear memory Hao et al. [2024]. Phi-GaLore differs
111 by operating directly on full gradients with learned and analytic subspaces and can be combined with
112 FLORA-like momentum compression at the optimizer-state level.

113 Randomized sketching and low-rank structure. Frequent Directions and related sketches track
114 evolving leading eigenspaces with sublinear memory, supporting efficient subspace estimation under
115 stochastic training Feinberg et al. [2023]. Theory and experiments on matrix sensing and factorization
116 reveal incremental low-rank learning and concentrated spectra Jin et al. [2023], Li et al. [2020],
117 motivating small-rank approximations in practice. Phi-GaLore leverages per-step CountSketch to
118 form an EMA $r \times r$ core and performs only a tiny SVD, independent of the full matrix size.

119 SciML transformers and neural operators. PINNsFormer reinterprets physics-informed learning
120 through sequence modeling Zhao et al. [2023]; Transolver introduces physics-aware attention for com-
121 plex geometries Wu et al. [2024]; position-induced transformers and universal physics transformers
122 explore scalable operator learning Chen and Wu [2024], Alkin et al. [2024]; graph transformer neural
123 operators emphasize modularity across geometries Bryutkin et al. [2024]. Molecular models such as
124 GeoMFormer and multi-modal transformers stress invariance and equivariance Chen et al. [2024],
125 Luo et al. [2022], Rong et al. [2020]. To our knowledge, existing low-rank projection methods for
126 training do not inject analytic physics bases into gradient subspaces, nor do they amortize projector
127 computation across layers with explicit cross-layer sharing tailored to SciML.

128 Comparison and applicability. We use GaLore as a baseline where projection is applicable Zhao et al.
129 [2024]. PEFT methods are not directly comparable in our setting because they limit update rank
130 and change the optimization path, which can undermine physics fidelity Guo et al. [2023], Liu et al.
131 [2024], Lialin et al. [2023], Pan et al. [2024], Biderman et al. [2024]. FLORA is complementary and
132 compatible with our optimizer wrapper Hao et al. [2024]. Phi-GaLore is designed for full-parameter
133 SciML training under memory constraints with explicit symmetry preservation, a niche not served by
134 current PEFT or GaLore variants.

135 3 Background

136 3.1 Problem setting and notation

137 Consider training a SciML model with parameter matrices $W \in \mathbb{R}^{m \times n}$. Let G denote the per-
138 step gradient with respect to W . Standard adaptive optimizers (e.g., AdamW) maintain two state

tensors per parameter, which becomes prohibitive at billion-parameter scale. Gradient low-rank projection seeks a projector $P \in \mathbb{R}^{m \times k}$, with $k \ll m$, and replaces G by the projected gradient $G_{\text{proj}} = P(P^\top G)$, constraining updates to the span of P while keeping the full parameter set trainable. Assuming the leading energy of G concentrates in a low-dimensional subspace, this projection can reduce the optimizer-state footprint by storing and updating only compressed statistics Zhao et al. [2024].

3.2 GaLore and limitations in SciML

GaLore constructs P via periodic full SVDs of large gradient matrices and yields substantial optimizer memory savings in NLP Zhao et al. [2024]. However, for wide SciML layers, SVD refreshes are costly, and the learned subspaces are agnostic to physics. In incompressible flows or equivariant molecular models, discarding gradient directions aligned with constraints (e.g., divergence-free directions or SO(3) mixing structure) can degrade fidelity.

3.3 Randomized sketching

Sketching compresses matrices by hashing rows and columns into a small number of buckets with random signs and aggregating values. CountSketch preserves leading subspace information with low overhead, and an exponential-moving-average (EMA) stabilizes subspace tracking under SGD noise. Forming an $r \times r$ covariance of the EMA core and computing a tiny SVD yields leading directions at negligible cost compared to full SVDs Feinberg et al. [2023].

3.4 Physics-aware subspaces

Many SciML tasks exhibit symmetries that can be encoded analytically. For incompressible flow, divergence-free bases derived from streamfunction modes align with the constraint $\nabla \cdot u = 0$. For molecular transformers, SO(3) structure suggests block bases aligned with irreducible representations to preserve equivariant mixing Chen et al. [2024]. Augmenting learned subspaces with such analytic bases helps ensure projected updates do not violate key invariances. This is coherent with physics-aware modeling trends in PINNsFormer and Transolver Zhao et al. [2023], Wu et al. [2024], and with scalable operator learning Chen and Wu [2024], Alkin et al. [2024], Bryutkin et al. [2024].

3.5 Low-rank structure in optimization

Empirical and theoretical studies show that training dynamics often concentrate in low-dimensional subspaces, supporting small ranks for efficient approximation Feinberg et al. [2023], Jin et al. [2023], Li et al. [2020]. This motivates working with r in the tens (e.g., 16, 32, 64) and focusing compute on $r \times r$ operations.

3.6 Assumptions

We assume that gradient energy maintains a low effective rank across training; that physics bases can be attached to layers with clear physical semantics (e.g., velocity heads) without semantic mismatch; and that layers at similar resolution or function can share subspaces in deep transformers. These assumptions align with recent SciML architectures and equivariant molecular models Alkin et al. [2024], Bryutkin et al. [2024], Chen et al. [2024].

4 Method

Phi-GaLore is a two-stage, symmetry-aware gradient projection framework with hierarchical sharing, implemented as a drop-in optimizer wrapper for AdamW and 8-bit variants.

4.1 Stage 1: Per-step randomized sketching with EMA

For each large parameter matrix $W \in \mathbb{R}^{m \times n}$ with gradient G , compute a left CountSketch $S_\ell(G) \in \mathbb{R}^{r \times n}$ by hashing rows into r buckets with random signs, then apply a right CountSketch to obtain a tiny core $\tilde{G} \in \mathbb{R}^{r \times r}$. Maintain an exponential-moving-average core $\tilde{G}_{\text{ema}} \leftarrow \alpha \tilde{G}_{\text{ema}} + (1 - \alpha) \tilde{G}$

with $\alpha \in (0, 1)$ to suppress stochastic noise. Parameters are grouped by a share key encoding layer resolution and matrix type; the first parameter encountered per group updates the sketch each step, and the others reuse the group’s sketch. The per-step cost is $\mathcal{O}((m + n)r)$ for large matrices and negligible memory beyond the $r \times r$ core.

4.2 Stage 2: Periodic tiny SVD and physics augmentation

Every T steps per group, form the covariance $C = \tilde{G}_{\text{ema}} \tilde{G}_{\text{ema}}^\top \in \mathbb{R}^{r \times r}$ and compute its SVD $C = U \Sigma U^\top$. Select the top k columns of U ($k \leq r$) to obtain the learned basis U_{sketch} in sketch space. Lift U_{sketch} back to row space using the inverse bucket mapping implied by CountSketch, producing $P_{\text{learned}} \in \mathbb{R}^{m \times k}$. Apply QR orthonormalization for numerical stability. If a physics basis $B \in \mathbb{R}^{m \times k_b}$ is registered for the group (e.g., divergence-free Fourier modes at a velocity head or SO(3)-aligned blocks for equivariant layers), concatenate and perform two-pass QR to produce the final projector $P \in \mathbb{R}^{m \times k'}$, where $k' \leq k + k_b$ after removing redundancy.

4.3 Gradient replacement and hierarchical sharing

Before each optimizer update, replace G with $G_{\text{proj}} = P(P^\top G)$ for matrices exceeding a size threshold; vectors and small matrices are left unmodified to avoid overhead dominance. The base optimizer then updates weights. To shrink the number of periodic SVDs and lifting operations, groups of layers that share resolution or function (e.g., attention value and output projections within the same depth bucket) share sketches, tiny SVDs, and physics bases. This amortizes compute across deep models and reduces both time and storage for projectors.

4.4 Compatibility, complexity, and storage

EMA cores and projectors are stored in reduced precision (e.g., BF16 or FP16), and the wrapper is compatible with 8-bit optimizer states. Because FLORA can be interpreted as gradient compression via random projections with resampling Hao et al. [2024], a per-group policy can, in principle, select between Phi-GaLore projection and FLORA-style compression to minimize overhead, though this switch was not activated in our short experiments. The per-step sketching cost is linear in $(m + n)$ with factor r , and the periodic cost per group is $\mathcal{O}(r^3)$ for the tiny SVD, independent of m and n . Storage is dominated by $r \times r$ EMA cores and the final projectors per group in reduced precision. In contrast, GaLore’s periodic full SVD on $m \times n$ gradients imposes quadratic or worse time in hidden size Zhao et al. [2024].

4.5 Design rationale and implementation

Randomized EMA sketches avoid the heavy SVDs central to GaLore yet track evolving subspaces under SGD noise Feinberg et al. [2023]. Physics-aware augmentation ensures projected updates cannot violate constraints that matter in SciML, aligning with invariance and equivariance principles in PDEs and molecules Zhao et al. [2023], Wu et al. [2024], Chen et al. [2024]. Hierarchical sharing leverages homogeneity across layers to reduce overhead. Implementation includes CountSketch for rows and columns, EMA maintenance, $r \times r$ covariance SVD, lifting from sketch to row space by unsketching with count normalization, QR orthonormalization in float32, gradient replacement for large matrices, and per-step timing. A registry maps parameters or groups to optional physics bases. Projection is skipped for matrices below a size threshold to prevent overhead dominance.

Phi-GaLore: per-step procedure with periodic refresh Inputs: gradients $\{G^{(\ell)}\}$, groups $\mathcal{G}$, rank r , basis size k , EMA factor α , refresh interval T each group $g \in \mathcal{G}$ first parameter in g encountered Compute two-sided CountSketch core: $\tilde{G}^{(g)} \leftarrow S_\ell(G^{(g)}) S_r^\top$ EMA update: $\tilde{G}_{\text{ema}}^{(g)} \leftarrow \alpha \tilde{G}_{\text{ema}}^{(g)} + (1 - \alpha) \tilde{G}^{(g)}$ step % $T = 0$ $C \leftarrow \tilde{G}_{\text{ema}}^{(g)} \tilde{G}_{\text{ema}}^{(g)\top}$; $[U, \Sigma] \leftarrow \text{SVD}(C)$ $U_{\text{sketch}} \leftarrow U[:, 1:k]$; lift to row space: $P_{\text{learned}} \leftarrow \text{Unsketch}(U_{\text{sketch}})$ Orthonormalize: $Q \leftarrow \text{QR}(P_{\text{learned}})$ Optional physics basis B ; concatenate $[Q, B]$ and re-orthonormalize to get $P^{(g)}$ each parameter matrix $W^{(\ell)}$ in group g $\text{size}(W^{(\ell)})$ above threshold Project gradient: $G^{(\ell)} \leftarrow P^{(g)}(P^{(g)\top} G^{(\ell)})$ Call base optimizer step (e.g., AdamW) on updated gradients

230 5 Experimental Setup

231 We evaluate Phi-GaLore on three short, controlled PyTorch experiments designed to instrument
232 gradient alignment, projection overhead, and physics fidelity. All runs use a single CUDA device
233 with mixed-precision compute dtype torch.float16, as reported by the logs. Given the brevity of these
234 runs, the goal is to validate mechanisms rather than to establish final accuracy or memory savings.

235 5.1 Experiment 1: Gradient low-rankness and projection efficiency

236 Models: (A) MLPField mapping grid inputs to velocity-like outputs; (B) a tiny transformer for
237 molecular-like regression. Methods: Full-rank AdamW; Phi-GaLore with rank $r = 32$; and a naive
238 GaLore replication with periodic SVD. Refresh intervals in this setup were $T = 50$ for Phi-GaLore
239 and $T = 20$ for GaLore (the latter incurred an initial SVD cost at step 1 and did not refresh again
240 within 8 steps). Measurements: per-step training loss; projection overhead via CUDA events in
241 milliseconds; cosine similarity between full gradients and Phi-GaLore-projected gradients on the
242 largest targeted parameter. Artifacts: training_loss_microbench.pdf, gradient_rank_microbench.pdf,
243 projection_overhead_microbench.pdf, alignment_phi-galore_microbench.pdf.

244 5.2 Experiment 2: Physics-aware subspace for incompressible Navier-Stokes

245 Task: a toy 2D incompressible Navier-Stokes setting with a model featuring a velocity output head.
246 Variants: Full-rank; Phi-GaLore without physics basis; Phi-GaLore with a divergence-free Fourier
247 basis injected at the output layer with $K = 16$ modes, orthonormalized. Measurements: training
248 loss per step; relative L2 error (rL2) under clean conditions, boundary-condition noise, and mesh
249 jitter; divergence norm under clean conditions; projection overhead in milliseconds. Artifacts:
250 training_loss_navierstokes.pdf, rL2_navierstokes.pdf, divergence_norm_navierstokes.pdf.

251 5.3 Experiment 3: Hierarchical projector sharing for toy molecular regression

252 Task: a toy molecular regression with a small transformer; robustness is assessed under random
253 rotations and Gaussian coordinate noise. Variants: Full-rank; Phi-GaLore with sharing ON (grouping
254 attention value and output projections by depth bucket); Phi-GaLore with sharing OFF; GaLore.
255 Measurements: training loss per step; projection overhead in milliseconds; mean absolute error
256 (MAE) under rotations and under coordinate noise. Artifacts: training_loss_molecular.pdf, projec-
257 tion_overhead_molecular.pdf, mae_molecular.pdf.

258 5.4 Baselines and rationale

259 GaLore is included where projection is applicable Zhao et al. [2024]. We do not include PEFT
260 baselines (e.g., LoRA family) because they restrict update rank and alter training dynamics, making
261 direct comparison to full-parameter projection less informative for physics fidelity Guo et al. [2023],
262 Liu et al. [2024], Lialin et al. [2023], Pan et al. [2024]. Our wrapper is compatible with 8-bit states
263 and FLORA-style compression, but these were not the primary focus of this verification pass Hao
264 et al. [2024].

265 5.5 Links to SciML literature

266 The chosen tasks probe core desiderata for operator-learning transformers and molecular models:
267 alignment of projected and full gradients, overhead behavior with and without sharing, and efficacy
268 of physics-aware bases for constraint preservation Zhao et al. [2023], Wu et al. [2024], Alkin et al.
269 [2024], Bryutkin et al. [2024], Chen et al. [2024].

270 6 Results

271 We report all observed metrics from the logs and include all generated figures, embedding each
272 exactly once with captions.

6.1 Experiment 1: Micro-bench alignment and overhead

Training loss trajectories over 8 steps were similar across methods. For MLPField, Full-rank loss ranged from 0.019340 to 0.014409; Phi-GaLore ranged from 0.020019 to 0.014409; GaLore ranged from 0.019637 with subsequent values around 0.015151-0.016893. Projection overhead for MLPField was 59.501 ms at step 1 for Phi-GaLore (initialization) then approximately 0.86 ms per step; GaLore incurred 33.917 ms at step 1 then approximately 0.24 ms per step. For the tiny transformer, Phi-GaLore incurred 6.147 ms at step 1 then approximately 2.63 ms per step; GaLore incurred 88.357 ms at step 1 then approximately 0.98 ms per step. Cosine similarity between full and Phi-GaLore-projected gradients on the largest parameter was approximately 1.0000 for both MLPField (head_vel.weight) and the transformer (blocks.0.qkv.weight), indicating near-perfect alignment at $r = 32$.

6.2 Experiment 2: Physics-aware basis on toy Navier-Stokes

Full-rank and Phi-GaLore without basis reduced training loss from 0.237570 to 0.120997 over 8 steps. With a divergence-free basis ($K = 16$), loss decreased from 0.237570 to 0.209511, indicating slower early convergence. Clean rL2 was 0.6198 for Full and for Phi (no basis) and 0.9320 for Phi with basis; under boundary-condition noise, rL2 was 0.6212 (Full), 0.6216 (Phi), and 0.9322 (Phi+B); under mesh jitter, 0.6415 (Full), 0.6415 (Phi), and 0.9335 (Phi+B). The divergence norm under clean conditions was $1.2319\text{e}-01$ for Full and Phi, and $5.7455\text{e}-02$ for Phi+B, about a 53% reduction. Projection overhead for Phi variants was approximately 0.86-0.97 ms per step after initialization.

6.3 Experiment 3: Hierarchical sharing for toy molecular regression

All methods achieved comparable training losses (mid-20s to high-20s) over 8 steps. Phi-GaLore with sharing reduced per-step overhead to approximately 3.58 ms from approximately 5.12 ms without sharing, a roughly 30% reduction. GaLore showed about 108.484 ms at step 1 and about 1.39 ms thereafter. Robustness MAE under rotations was 25.0783 (Full), 24.9510 (Phi-share, best), 25.2917 (Phi-noshare), and 25.4712 (GaLore). Under coordinate noise, MAE was 25.1290 (Full), 25.8195 (Phi-share), 25.3672 (Phi-noshare), and 25.1745 (GaLore).

6.4 Fairness, limitations, and scope

These are very short runs (8 steps), so periodic refreshes did not recur, and steady-state overheads could not be measured. No memory profiling was performed, so optimizer-state savings are not quantified here. The physics basis was applied only at the velocity head with $K = 16$ and may require tuning. Despite these limits, the experiments confirm: (i) very high gradient alignment at $r = 32$, (ii) meaningful divergence reduction via physics-aware bases, and (iii) significant overhead reduction from hierarchical sharing. In longer runs, GaLore’s repeated SVD cost would recur; our design targets that regime by relying on $r \times r$ SVDs only Zhao et al. [2024].

7 Conclusion

We presented Physically-Aware Hierarchical GaLore (Phi-GaLore), a two-stage, symmetry-aware gradient projection framework for SciML. Phi-GaLore replaces full-matrix SVDs with a tiny $r \times r$ SVD on randomized EMA sketches, augments learned subspaces with analytic physics bases, and amortizes computation via hierarchical projector sharing. Implemented as a PyTorch optimizer wrapper, it preserves full-parameter training and is compatible with 8-bit states and FLORA-style momentum compression Zhao et al. [2024], Hao et al. [2024].

Across three short, instrumented studies, Phi-GaLore achieved near-perfect cosine alignment to full gradients at rank 32, substantially reduced divergence norms on a toy incompressible Navier-Stokes task when injecting a divergence-free basis, and reduced per-step projection overhead by roughly 30% via hierarchical sharing. These brief runs did not capture repeated SVD refresh cycles for GaLore, so steady-state overhead comparisons and memory savings at scale remain to be demonstrated. The divergence-free basis also introduced an early accuracy-constraint trade-off, suggesting the need to tune basis size and placement.

Future work will scale to long-horizon (approximately 150k-step) training on PINNsFormer, Transolver, and molecular transformers under fixed GPU budgets to quantify optimizer-state memory, throughput, and accuracy, and to test SO(3) bases for equivariant layers Zhao et al. [2023], Wu et al. [2024], Chen et al. [2024], Alkin et al. [2024], Bryutkin et al. [2024]. We will ablate rank r , refresh interval T , and basis size K to balance physics fidelity and downstream error, and explore per-group policies that select between Phi-GaLore projection and FLORA-style compression to minimize overhead Hao et al. [2024]. By aligning gradient compression with physical symmetries and amortized computation, Phi-GaLore aims to unlock full-parameter optimization of billion-scale SciML models on commodity GPUs.

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Figure 1: Training loss on micro-bench models (MLPField and tiny transformer) across methods (filename: training_loss_microbench.pdf)

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Figure 2: Gradient stable-rank diagnostics on micro-bench snapshots (filename: gradient_srank_microbench.pdf)

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Figure 3: Projection overhead on micro-bench per step and method (filename: projection_overhead_microbench.pdf)

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Figure 4: Cosine alignment of Phi-GaLore vs. full gradients on largest parameters (filename: alignment_phi-galore_microbench.pdf)

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Figure 5: Training loss on toy Navier-Stokes across variants (filename: training_loss_navierstokes.pdf)

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Figure 6: Relative L2 error on Navier-Stokes under clean, boundary-noise, and jitter conditions (filename: rL2_navierstokes.pdf)

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Figure 7: Divergence norm on Navier-Stokes under clean conditions (filename: divergence_norm_navierstokes.pdf)

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Figure 8: Training loss on toy molecular regression across methods (filename: training_loss_molecular.pdf)

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Figure 9: Projection overhead on toy molecular regression with and without sharing (filename: projection_overhead_molecular.pdf)

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Figure 10: Robustness MAE on molecular task under rotations and coordinate noise (filename: mae_molecular.pdf)